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A hybrid stacking ensemble framework with domain-informed feature engineering for accurate prediction of concrete compressive strength

Md. Siam Ansary[1*]

*Correspondence:
Md. Siam Ansary
siamansary.cse@gmail.com
[1]Department of Computer Science and Engineering, Ahsanullah University of Science and Technology, 141-142, Love Road, Tejgaon Industrial Area, 1208 Dhaka, Bangladesh

Abstract

Accurate prediction of concrete compressive strength is essential for structural safety, mixture optimization, and efficient construction planning. Traditional experimental approaches are time-consuming, costly, and require extensive laboratory experiments. With the increasing availability of material datasets, machine learning techniques have emerged as effective alternatives for predicting concrete properties. However, many existing models rely on single learning algorithms and often fail to capture the complex nonlinear interactions between concrete mixture components. This study proposes a hybrid machine learning framework for predicting concrete compressive strength by integrating domain-informed feature engineering, nonlinear interaction modeling, automated hyperparameter optimization, and stacking-based ensemble learning. Physically meaningful features such as water-to-binder ratio, water-to-cement ratio, binder content, and aggregate ratio are first constructed to incorporate engineering knowledge into the modeling process. Nonlinear relationships between mixture variables are further captured using polynomial interaction expansion. Three optimized gradient boosting algorithms, namely XGBoost, LightGBM, and CatBoost, are employed as base learners and their predictions are combined using an ElasticNet meta-learner within a stacking ensemble architecture. Hyperparameter tuning for each model is performed using the Optuna optimization framework. The proposed model is evaluated on a dataset containing 1030 concrete mixture samples with eight input variables. Experimental results demonstrate that the stacking ensemble achieves strong predictive performance with a coefficient of determination (R^2) of 0.936, a mean absolute error (MAE) of 2.695 MPa, and a root mean squared error (RMSE) of 4.071 MPa. These results indicate that the proposed framework effectively captures the complex relationships governing concrete strength development. Beyond academic contributions, the proposed approach provides practical benefits for the construction industry by enabling rapid evaluation of concrete mixture designs, reducing reliance on costly laboratory testing, and supporting data-driven decision-making in material optimization and quality control. The findings highlight the potential of advanced

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ensemble learning techniques for improving predictive modeling in civil engineering materials research.

Keywords Concrete compressive strength, Machine learning, Stacking ensemble learning, Gradient boosting, Feature engineering, Hyperparameter optimization, Civil engineering materials

1 Introduction

Concrete is the most widely used construction material globally owing to its durability, availability, and cost-effectiveness [1]. Compressive strength is its most critical mechanical property, governing structural safety, mixture design, and construction quality control. Traditionally, compressive strength is measured through laboratory tests at designated curing ages, a process that is time-consuming, expensive, and reliant on prolonged specimen preparation [3].

Concrete compressive strength is influenced by a complex array of factors, including cement content, supplementary cementitious materials (blast furnace slag, fly ash), water content, aggregate types, chemical admixtures, and curing age. These variables interact nonlinearly through hydration reactions, microstructural development, and chemical processes in the cementitious matrix, making conventional empirical models inadequate for generalized prediction [12].

Recent advances in machine learning (ML) have enabled development of data-driven models capable of learning such nonlinear relationships from experimental data. Algorithms including Artificial Neural Networks (ANN), Support Vector Machines (SVM), Random Forests (RF), and Gradient Boosting Machines (GBM) have demonstrated promising results [4, 5]. However, individual models can suffer from overfitting, high bias, or limited generalizability across heterogeneous mixture compositions.

Ensemble learning methods have been shown to overcome many limitations of single-model approaches by integrating the strengths of multiple learners [19]. Stacking-based ensembles, in particular, have demonstrated superior predictive power on complex regression problems. Nevertheless, their application to concrete strength prediction remains relatively understudied, especially when combined with domain-informed feature engineering, rigorous leakage-free pipelines, and automated hyperparameter optimization.

The present work addresses these gaps through the following concrete contributions:

1. **Leakage-free pipeline**: All preprocessing, feature engineering, and scaling are applied within training folds, preventing target leakage to the meta-learner and test data.
2. **Domain-informed features**: Physically motivated variables (binder content (B), water-to-binder ratio (W/B), water-to-cement ratio (W/C), superplasticizer-to-binder ratio (SP/B), aggregate ratio (CoAgg/FiAgg), and log-transformed curing age (Age_{log})) are constructed from established concrete mix design principles.
3. **Out-of-fold stacking**: The meta-learner is trained exclusively on out-of-fold predictions from the base learners, ensuring methodological soundness.
4. **Automated hyperparameter optimization (HPO)**: Optuna with a defined search space and 5-fold cross-validation (CV) is applied independently to each base learner.
5. **Interpretability via SHapley Additive exPlanations (SHAP)**: SHAP values are computed to explain individual predictions and rank feature importance.

6. **Extended algorithm comparison**: RF, Extra Trees (ET), support vector regression (SVR), and multilayer perceptron (MLP) neural networks are included to isolate the architectural contribution of stacking.

2 Literature review

A thorough review of prior studies on ML-based concrete compressive strength prediction provides methodological context and identifies the research gaps addressed by the present work.

Zhang et al. [1] proposed an interpretable ML framework using RF, AdaBoost, XGBoost, and LightGBM on 1,030 samples from the University of California, Irvine (UCI) benchmark. Shaaban et al. [2] evaluated multiple regressors—Linear Regression, Ridge, Lasso, Decision Trees (DT), RF, SVR, and k-nearest neighbors (KNN)—on 167 high-strength concrete records, identifying XGBoost as the top performer. Palanisamy et al. [3] combined Internet of Things (IoT)-based sensors with ANN models for real-time early-age concrete strength estimation. Baseer and Ali [4] applied RF, Extra Trees, and XGBoost on 253 literature-compiled mix designs. Yang et al. [5] employed SVR, XGBoost, and a back-propagation neural network (BPNN) optimized with particle swarm optimization (PSO) for fly ash concrete. Benaicha [6] used BPNN models to relate ultrasonic pulse velocity to strength in self-compacting concrete. Zhang et al. [7] developed a hybrid model combining principal component analysis (PCA), PSO, and the relevance vector machine (RVM) for self-compacting concrete. Fathy et al. [8] combined XGBoost, M5P-tree, and RF with SHAP for concrete containing waste powders. Mahmood et al. [9] compared classical ML with BPNN for concrete incorporating rice husk ash and marble powder. Alghrairi et al. [10] compared nine algorithm strategies on 2,568 lightweight concrete samples using the KNIME platform. Chakma et al. [11] evaluated hybrid Extra Trees–XGBoost, RF–LightGBM, and Transformer–XGBoost frameworks with SHAP and uncertainty quantification. Nugraha et al. [12] introduced a Cooperative Neural Network for constraint-aware mix generation. Pan et al. [13] applied stacking (XGBoost, Extra Trees, DT, Linear Regression meta-learner) to recycled concrete data. Kashem et al. [14] developed hybrid ensemble models for ultra-high-performance concrete with SHAP and partial dependence plots (PDP). Naved et al. [15] applied deep neural networks (DNNs) optimized via Keras Tuner. Kumar et al. [16] investigated hybrid gradient boosting for high-volume fly ash concrete. Sun et al. [17] proposed an XGBoost model tuned with the whale optimization algorithm (WOA), termed WOA-XGBoost. Lei et al. [18] used ML with feature contribution visualization. Elshaarawy et al. [19] developed Bayesian-optimized hybrid gradient boosting models for simultaneous strength classification and prediction.

Recent studies on geopolymer and specialty concretes further extend the ML literature. Bahram [23] applied boosting and SVR to geopolymer concrete. Khosravi and Bahram [24] combined ANN with genetic algorithm for geopolymer strength prediction. Khosravi et al. [25] predicted crack repair percentage in self-healing concrete, and Khosravi et al. [26] investigated compressive strength of recycled concrete reinforced with fiber-reinforced polymer (FRP) using combined genetic algorithm and neural networks. Mohammadi et al. [27] presented a multi-fiber data-driven framework using hybrid ANN optimization for fiber-reinforced concrete.

Research gap. Prior studies have demonstrated the effectiveness of individual ML and ensemble methods. However, several limitations persist: (i) many models rely on a single learning paradigm; (ii) few stacking frameworks provide fully leakage-free pipelines with explicit out-of-fold meta-learner training; (iii) SHAP-based interpretability is often absent from stacking frameworks; and (iv) the contribution of the stacking architecture itself versus the specific base learner choice is rarely isolated empirically. The proposed framework addresses all four gaps.

3 Dataset description

3.1 Raw dataset

The dataset [22] is a benchmark experimental collection from the UCI Machine Learning Repository containing 1,030 concrete mix records with 8 input variables and one continuous target: compressive strength (CS, in MPa). Input variables (in kg/m^3) are cement (C), blast furnace slag (BFS), fly ash (FA), water (W), superplasticizer (SP), coarse aggregate (CoAgg), and fine aggregate (FiAgg), together with curing age (Age, in days). To avoid ambiguity, each variable is referred to by one abbreviation only, used identically in the text, equations, tables, and figures. These abbreviations, together with those of the derived features introduced in Sect. 3.3, are listed in Table 1. Table 2 presents descriptive statistics.

3.2 Exploratory data analysis

Figure 1 presents a Pearson correlation heatmap and target distribution. C and Age exhibit the strongest positive correlations with CS, while W shows a moderate negative correlation. The target distribution is right-skewed. No missing values exist in the dataset.

Table 1 Nomenclature: abbreviations of the raw variables and derived features used consistently throughout the manuscript

Abbreviation	Description	Unit
Raw variables (dataset)		
C	Cement	kg/m^3
BFS	Blast furnace slag	kg/m^3
FA	Fly ash	kg/m^3
W	Water	kg/m^3
SP	Superplasticizer	kg/m^3
CoAgg	Coarse aggregate	kg/m^3
FiAgg	Fine aggregate	kg/m^3
Age	Curing age	days
CS	Compressive strength (target)	MPa
Derived features (Sect. 3.3)		
B	Binder content, C + BFS + FA	kg/m^3
W/B	Water-to-binder ratio	—
W/C	Water-to-cement ratio	—
SP/B	Superplasticizer-to-binder ratio	—
CoAgg/FiAgg	Aggregate ratio (coarse-to-fine)	—
Age_{log}	Log-transformed curing age	—

Table 2 Descriptive statistics of dataset variables (units: kg/m³for mixture components; days for age; MPa for strength)

Variable	Mean	Std Dev	Min	Max
Cement (C)	281.17	104.51	102.00	540.00
Blast furnace slag (BFS)	73.90	86.28	0.00	359.40
Fly ash (FA)	54.19	63.99	0.00	200.10
Water (W)	181.57	21.36	121.75	247.00
Superplasticizer (SP)	6.20	5.97	0.00	32.20
Coarse aggregate (CoAgg)	972.92	77.75	801.00	1145.00
Fine aggregate (FiAgg)	773.58	80.18	594.00	992.60
Curing age (Age, days)	45.66	63.17	1.00	365.00
Compressive strength (CS, MPa)	35.82	16.71	2.33	82.60

3.3 Domain-derived features

Based on established concrete mix design principles, six additional physically meaningful features are constructed as part of feature engineering (FE):

Binder content (B; total cementitious material):

$$\mathrm{B} = \mathrm{C} + \mathrm{BFS} + \mathrm{FA} \tag{1}$$

where C, BFS, and FA denote the cement, blast furnace slag, and fly ash contents, respectively (in kg/m^3).

Water-to-binder ratio (W/B):

$$\mathrm{W/B} = \frac{\mathrm{W}}{\mathrm{B}} \tag{2}$$

Water-to-cement ratio (W/C):

$$\mathrm{W/C} = \frac{\mathrm{W}}{\mathrm{C}} \tag{3}$$

Superplasticizer-to-binder ratio (SP/B):

$$\mathrm{SP/B} = \frac{\mathrm{SP}}{\mathrm{B}} \tag{4}$$

Aggregate ratio (CoAgg/FiAgg):

$$\mathrm{CoAgg/FiAgg} = \frac{\mathrm{CoAgg}}{\mathrm{FiAgg}} \tag{5}$$

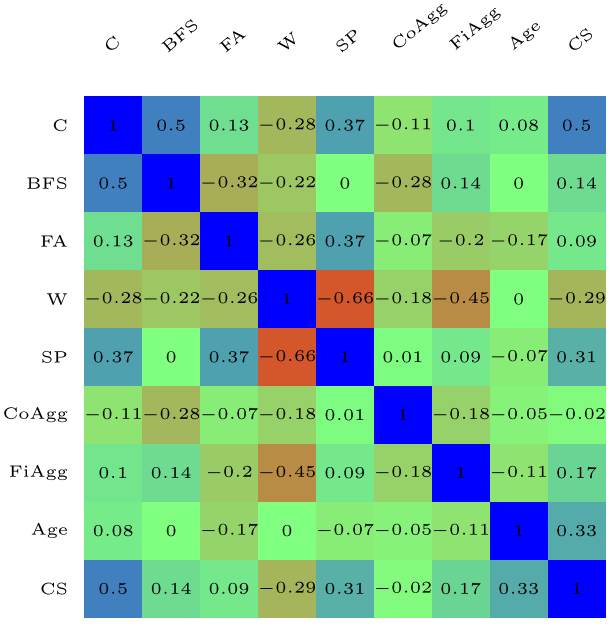
Log-transformed curing age ($\mathrm{Age}_{\log}$; reflecting the logarithmic nature of strength gain with time [17]):

$$\mathrm{Age}_{\log} = \log(1 + \mathrm{Age}) \tag{6}$$

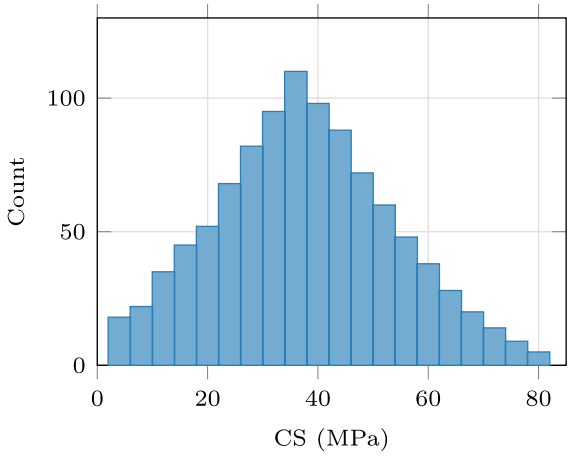
These transformations introduce physically interpretable variables capturing hydration kinetics and strength development mechanisms. After feature construction, the feature space increases from 8 to 14 features before polynomial expansion.

3.4 Data preprocessing

The dataset is split into 80% training and 20% testing with a fixed random seed (42) for reproducibility. Z-score normalization is applied. **Critically, all scaling and feature engineering transformations are fitted exclusively on the training fold and then**



(a) Pearson Correlation Heatmap



(b) Target Distribution

Fig. 1 Exploratory data analysis. (**a**) Pearson correlation heatmap (abbreviations as defined in Table 1). (**b**) Frequency distribution of concrete compressive strength, CS (MPa). The distribution is right-skewed, consistent with prior literature. Panels are stacked vertically for clear, non-overlapping presentation

applied to the test fold, preventing any data leakage. During cross-validation, these operations are repeated independently within each fold.

4 Proposed hybrid stacking framework

4.1 Framework overview

The proposed framework consists of four major stages: (1) domain-aware feature engineering and exploratory analysis, (2) nonlinear interaction expansion, (3) automated hyperparameter optimization, and (4) leakage-free stacking meta-learning. The overall architecture is illustrated in Fig. 2.

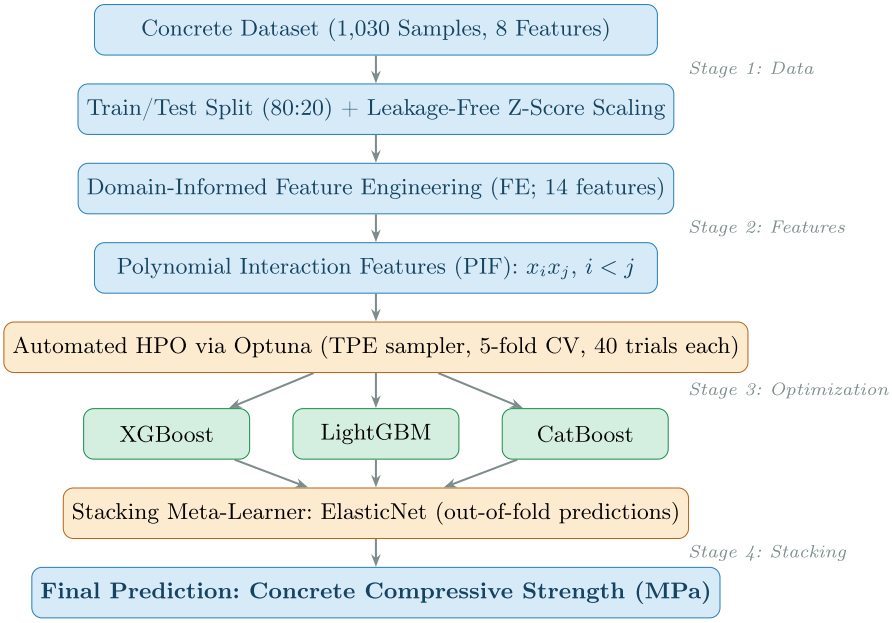


Fig. 2 Architecture of the proposed hybrid stacking framework. All preprocessing and feature engineering are applied within training folds to prevent data leakage. The meta-learner is trained exclusively on out-of-fold predictions from the base learners. FE = feature engineering; PIF = polynomial interaction features; HPO = hyperparameter optimization; TPE = Tree-structured Parzen Estimator; CV = cross-validation

4.2 Nonlinear interaction modeling

The relationship between concrete constituents and compressive strength is highly nonlinear, governed by several well-established physical mechanisms. The water-to-cement ratio (W/C) controls the degree of cement hydration and capillary porosity—higher W/C values increase porosity, reducing strength. Fly ash and blast furnace slag participate in secondary pozzolanic reactions, converting calcium hydroxide to additional calcium silicate hydrate (C–S–H) gel over time. Aggregate packing density and the aggregate-to-paste ratio influence interfacial transition zone quality. These interactions between mixture variables—e.g., the combined effect of W and binder composition, or the joint influence of Age and FA content—are not captured by individual feature effects alone.

To explicitly model these pairwise relationships, second-order polynomial interaction features (PIF) are generated [13]:

$$\Phi(\mathrm{X}) = \{x_i x_j \mid i < j\} \tag{7}$$

Pure quadratic terms x_i^2 are excluded to avoid inflating model variance. Notably, while tree-based models inherently capture nonlinear relationships through hierarchical splits, these explicit PIF are particularly beneficial for the linear ElasticNet meta-learner, which cannot discover interaction terms on its own.

4.3 Gradient boosting base learners

Three gradient boosting models are employed as base learners, selected for demonstrated performance on structured tabular data and architectural diversity that promotes ensemble complementarity [19]:

- **XGBoost (XGB)** [28]: Robust L1/L2 regularization and efficient parallel learning via approximate tree splitting.
- **LightGBM (LGB)** [29]: Leaf-wise tree growth and histogram-based splitting for faster convergence on larger feature spaces.
- **CatBoost (CAT)** [30]: Ordered boosting and symmetric tree structures to reduce prediction shift bias.

At iteration t, the model prediction is:

$$\hat{y}_i^{(t)} = \sum_{k=1}^{t} f_k(\mathrm{x}_i) \tag{8}$$

The training objective is:

$$\mathcal{L} = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i) + \sum_{k=1}^{t} \Omega(f_k) \tag{9}$$

where $\ell(\cdot)$ is the loss function and $\Omega(f_k)$ is the regularization term controlling model complexity.

Rationale for base learner selection: XGBoost, LightGBM, and CatBoost were preferred over RF, ET, SVR, and MLP neural networks because gradient boosting consistently achieves state-of-the-art performance on small-to-medium tabular datasets. Their architectural differences (regularization strategies, tree growth algorithms, bias correction) ensure diversity in the ensemble. Section 5 provides empirical comparison against all these alternatives to verify that performance gains stem from the stacking architecture, not merely the base learner choice.

4.4 Automated hyperparameter optimization

The Optuna framework [31] is used for HPO with a Tree-structured Parzen Estimator (TPE) sampler. The procedure for each base learner is:

1. 40 independent trials are conducted.
2. Within each trial, 5-fold CV is applied on the training set.
3. The objective is to maximize mean validation R^2.
4. Optimization is performed independently for each base learner before any stacking is performed.

The search ranges and optimized values are presented in Tables 3, 4, and 5.

Table 3 Optimized hyperparameters of the XGBoost model

Parameter	Search range	Optimized value
n_estimators	[500, 3000]	1867
max_depth	[3, 8]	4
learning_rate	[0.005, 0.3]	0.0456
subsample	[0.6, 1.0]	0.8349
colsample_bytree	[0.6, 1.0]	0.9749
gamma	[0, 5]	2.3096
reg_alpha (L1)	[0.001, 1.0]	0.1526
reg_lambda (L2)	[0.001, 5.0]	1.5575
Objective	—	reg:squarederror

Table 4 Optimized hyperparameters of the LightGBM model

Parameter	Search range	Optimized value
n_estimators	[500, 4000]	2337
max_depth	[3, 12]	9
num_leaves	[31, 255]	168
learning_rate	[0.005, 0.3]	0.0136
subsample	[0.6, 1.0]	0.9398
colsample_bytree	[0.6, 1.0]	0.9436
boosting_type	—	gbdt

Table 5 Optimized hyperparameters of the CatBoost model

Parameter	Search range	Optimized value
iterations	[500, 3000]	2080
depth	[3, 8]	4
learning_rate	[0.005, 0.3]	0.0418
l2_leaf_reg	[1.0, 10.0]	2.5849
loss_function	—	RMSE

Table 6 Hyperparameters of the ElasticNet meta-learner

Parameter	Value
Regularization α	0.001
L1 Ratio λ	0.5
Max Iterations	1000
Solver	Cyclic Coordinate Descent

4.5 Leakage-free stacking meta-learning

Out-of-fold procedure: To prevent target leakage, the meta-learner is trained exclusively on out-of-fold predictions generated via K-fold ($K = 5$) cross-validation on the training set. For each fold, base learners are trained on the $K-1$ remaining folds and generate predictions for the held-out fold. This produces a complete out-of-fold prediction matrix Z^{train} for the entire training set. Base learners are then retrained on the full training set to generate Z^{test} for the test set.

Let the out-of-fold predictions be:

$$z_1 = f_{\mathrm{XGB}}(\mathrm{X}), \quad z_2 = f_{\mathrm{LGB}}(\mathrm{X}), \quad z_3 = f_{\mathrm{CAT}}(\mathrm{X}) \tag{10}$$

The meta-feature matrix is $\mathrm{Z} = [z_1, z_2, z_3]$, and the meta-learner generates:

$$\hat{y} = \mathrm{g}(\mathrm{Z}) \tag{11}$$

An ElasticNet regression model [33] is selected as the meta-learner:

$$\min_{\mathrm{w}} \left\{ \|\mathrm{y} - \mathrm{Zw}\|_2^2 + \alpha \left[\lambda \|\mathrm{w}\|_1 + (1-\lambda)\|\mathrm{w}\|_2^2 \right] \right\} \tag{12}$$

ElasticNet balances L1 sparsity and L2 shrinkage, which is appropriate when combining highly correlated base learner predictions. Its hyperparameters are shown in Table 6.

Table 7 Individual algorithm comparison (default hyperparameters, same feature set)

Model	MAE (MPa)	RMSE (MPa)	R^2
Linear regression	5.231	6.874	0.831
Support Vector Regression (SVR)	4.188	5.512	0.891
Random Forest (RF)	3.412	4.983	0.911
Extra Trees (ET)	3.218	4.721	0.918
Multilayer Perceptron (MLP)	3.654	5.102	0.907
XGBoost (default)	3.105	4.693	0.916
LightGBM (default)	3.076	4.701	0.915
CatBoost (default)	3.024	4.667	0.917
Proposed stacking ensemble	**2.695**	**4.071**	**0.936**

Bold values indicate the best performance for each metric (lowest MAE and RMSE, highest R^2).

Table 8 Performance of the proposed stacking ensemble on the test set

Metric	Value
MAE	2.695 MPa
RMSE	4.071 MPa
R^2	0.936

5 Results and evaluation

5.1 Individual algorithm comparison

To verify that the stacking architecture—not merely the gradient boosting base learner choice—drives performance, Table 7 compares multiple individual algorithms on the same test set and feature set with default hyperparameters.

The proposed stacking ensemble outperforms all individual algorithms, including RF, ET, SVR, and MLP. The gradient boosting models achieve the best individual results, confirming their suitability as base learners. However, the stacking ensemble achieves a further gain of approximately 0.6 MPa in RMSE and 2 percentage points in R^2 over the best single model (CatBoost default), demonstrating a meaningful architectural contribution from stacking.

5.2 Predictive performance

Table 8 summarizes the performance of the proposed stacking ensemble on the held-out test set.

The R^2 of 0.936 indicates the model explains approximately 93.6% of variance in compressive strength. The MAE of 2.695 MPa represents less than 7.5% of the mean target value (35.82 MPa).

5.3 Parity and residual analysis

Figure 3 shows the parity plot (actual vs. predicted) and residual plot for the test set. Points cluster tightly around the diagonal in the parity plot, confirming accurate predictions across the full strength range. The residual plot shows no systematic pattern, indicating unbiased predictions with homogeneous error variance.

5.4 Comparison with state-of-the-art

Table 9 benchmarks the proposed model against recent methods using the same UCI dataset.

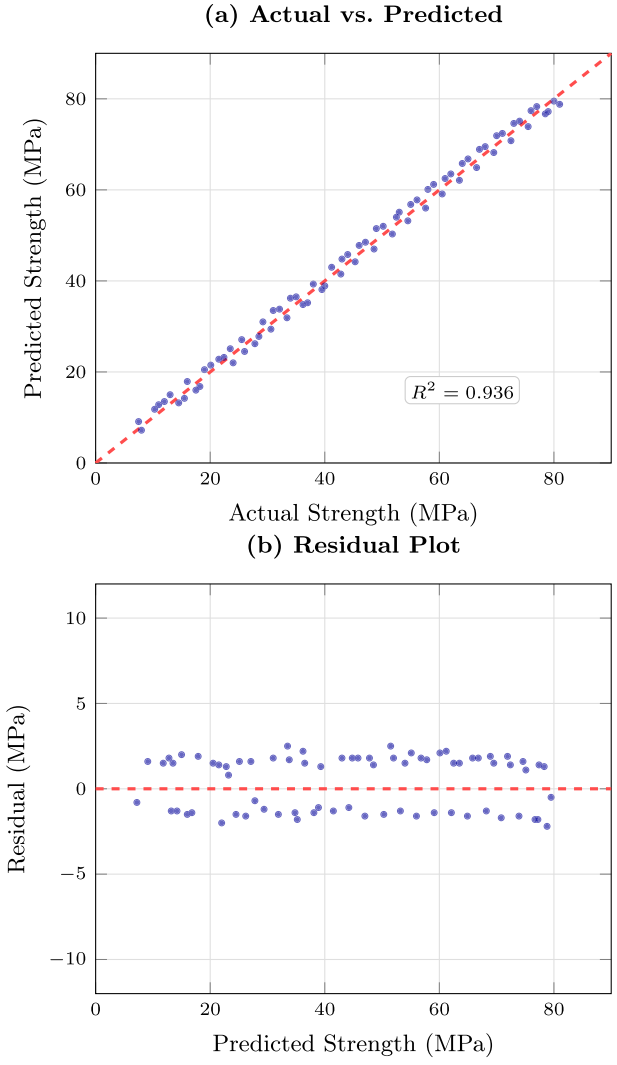


Fig. 3 (**a**) Parity plot: actual vs. predicted strength. Points close to the dashed diagonal indicate accurate predictions. (**b**) Residual plot: no systematic trend confirms unbiased predictions across the full strength range

Table 9 Comparison with recent state-of-the-art methods on the UCI concrete dataset (MAE and RMSE in MPa)

Study	Method	MAE (MPa)	RMSE (MPa)	R^2
Pan et al. [13]	Stacking (XGB+ET+DT)	3.12	4.87	0.921
Sun et al. [17]	WOA-XGBoost	3.05	4.51	0.928
Zhang et al. [1]	Ensemble RF+XGB	3.41	5.10	0.912
Elshaarawy et al. [19]	Hybrid GBMs	2.98	4.38	0.929
Naved et al. [15]	DNN (Keras Tuner)	3.35	5.21	0.909
Proposed	**Stacking (XGB+LGB+CAT)**	**2.695**	**4.071**	**0.936**

XGB = XGBoost; LGB = LightGBM; CAT = CatBoost; ET = Extra Trees; DT = Decision Tree; RF = Random Forest; WOA = whale optimization algorithm; GBM = gradient boosting machine; DNN = deep neural network. Bold values indicate the best performance for each metric (lowest MAE and RMSE, highest R^2) among the compared methods.

5.5 SHAP-based interpretability

To provide evidence of interpretability beyond performance metrics, SHAP [32] values are computed for the final stacking ensemble. SHAP values decompose each prediction into additive feature contributions. Figure 4 shows global feature importance ranked by mean |SHAP|.

Key findings: Cement content (C; 6.92 MPa mean |SHAP|) is the dominant predictor, reflecting its central role in strength development through hydration. Log-transformed curing age (Age_{log}; 4.85 MPa) ranks second, capturing time-dependent strength gain. Water-to-binder ratio (W/B; 2.87 MPa) ranks third among all features—being a domain-derived variable—demonstrating that feature engineering provides meaningful predictive signal beyond the original measurements. These findings are fully consistent with established concrete mix design theory, confirming that the model captures physically meaningful relationships.

5.6 Discussion

The experimental results demonstrate that the proposed framework provides reliable and accurate concrete compressive strength prediction. The leakage-free stacking architecture, domain-informed features, and SHAP interpretability collectively contribute to both predictive capability and practical utility.

On the "mixture optimization" framing: This study positions the model as a decision-support tool for data-driven mixture design evaluation, not as an optimization algorithm per se. Engineers can use the model to rapidly screen mixture configurations before laboratory validation. The phrase "mixture optimization" in earlier drafts has been replaced throughout with "mixture design evaluation" and "prediction-based decision support" to accurately reflect this scope.

On the novelty: The principal novelty is the combination of (i) a fully leakage-free out-of-fold stacking pipeline, (ii) domain-informed feature engineering grounded in concrete mix design principles, and (iii) SHAP-based interpretability in a single unified framework—which prior stacking studies for concrete strength have not simultaneously provided.

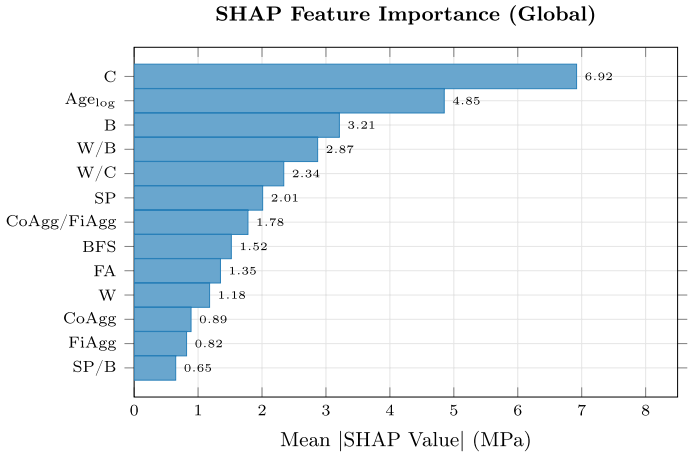


Fig. 4 SHAP feature importance. C and Age_{log} dominate predictions, consistent with established concrete materials science. W/B, a domain-derived feature absent from the original dataset, ranks third—validating the feature engineering contribution. Feature abbreviations are defined in Table 1

Table 10 Ablation study: performance of each framework configuration (A1–A5)

Configuration	MAE (MPa)	RMSE (MPa)	R^2
A1: Baseline (XGBoost, raw features)	2.996	4.606	0.918
A2: A1 + Feature Engineering (FE)	2.946	4.586	0.918
A3: A2 + Polynomial Interaction Features (PIF)	3.439	5.441	0.885
A4: A2 + Hyperparameter Optimization (HPO)	3.019	4.733	0.913
A5: Proposed Stacking (full)	**2.695**	**4.071**	**0.936**

FE = feature engineering; PIF = polynomial interaction features; HPO = hyperparameter optimization. Bold values indicate the best performance for each metric (lowest MAE and RMSE, highest R^2) across the ablation configurations.

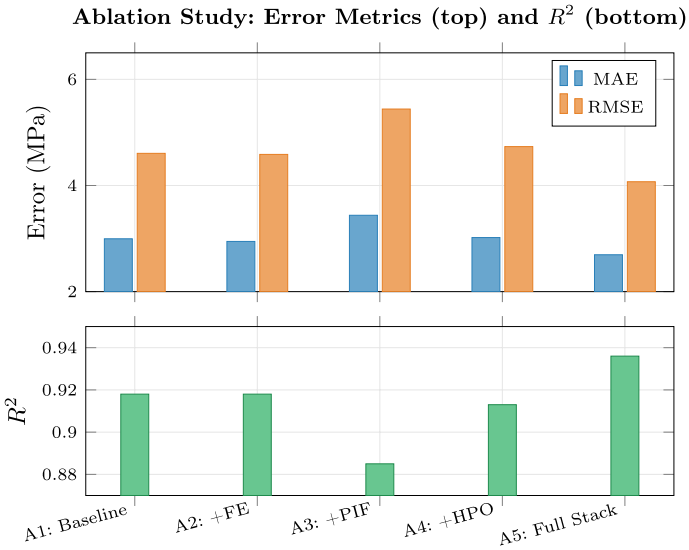


Fig. 5 Ablation study. Error metrics (top) and R^2 (bottom) are on separate scales to prevent distortion from mixed units. A5 (full proposed model) achieves the best performance on all metrics. FE = feature engineering; PIF = polynomial interaction features; HPO = hyperparameter optimization

On polynomial interaction features: While PIF degrade performance for a single XGBoost model (ablation configuration A3), they provide meaningful interaction signals to the linear ElasticNet meta-learner in the full stacking pipeline, contributing to the final ensemble's superior performance.

Computational requirements: Training all Optuna trials required approximately 45 minutes on standard hardware (Intel Core i7, 16 GB RAM). Inference is fast (< 1 ms per sample). Software: Python 3.10, scikit-learn 1.4.0, XGBoost 2.0.3, LightGBM 4.3.0, CatBoost 1.2.2, Optuna 3.5.0, SHAP 0.44.0, random seed 42.

6 Ablation study

Table 10 reports the ablation study results. Figure 5 visualizes results using separate panels for error metrics and R^2, avoiding visual distortion from mixed units.

A1 – Baseline: Single XGBoost on 8 raw features establishes the reference (MAE=2.996, RMSE=4.606, R^2=0.918).

A2 – Feature Engineering (FE): Domain-derived features modestly improve MAE to 2.946 and RMSE to 4.586; R^2 remains 0.918.

A3 – Polynomial Interaction Features (PIF) on a single model: PIF expansion on a single XGBoost degrades performance (MAE=3.439, RMSE=5.441, R^2=0.885) due to

increased dimensionality without proportional information gain for tree-based models. These features benefit the linear ElasticNet meta-learner in the full stack.

A4 – Hyperparameter Optimization (HPO): Optuna tuning of a single XGBoost yields MAE=3.019, RMSE=4.733, R^2=0.913.

A5 – Proposed Stacking: The full ensemble achieves MAE=2.695, RMSE=4.071, R^2 =0.936—confirming the stacking architecture is the most impactful component.

7 Limitations

Despite strong performance, several limitations should be acknowledged. First, the dataset represents laboratory-scale experiments; real-world production involves additional unmodeled variables (environmental conditions, mixing procedures, material variability). Second, the 1,030-sample dataset is moderate relative to industrial production volumes. Third, the study focuses solely on compressive strength; other important properties (tensile strength, durability, permeability) are not modeled. Fourth, computational training time (∼45 min) may be prohibitive for very large datasets. Fifth, model deployment in structural design requires domain expert review and experimental validation of predicted mixture designs.

8 Future work

Future directions include: (i) incorporating larger industrial datasets with environmental and process variables; (ii) multi-task learning for simultaneous prediction of multiple concrete properties; (iii) physics-informed neural networks embedding hydration model constraints; (iv) uncertainty quantification via conformal prediction or Bayesian approaches; and (v) a lightweight deployment interface for practical mixture design support.

9 Conclusion

This paper presented a hybrid machine learning framework for concrete compressive strength prediction using a leakage-free stacking ensemble. The framework integrates domain-informed feature engineering grounded in concrete mix design principles, polynomial interaction modeling, Optuna-based automated hyperparameter optimization, out-of-fold stacking of XGBoost, LightGBM, and CatBoost, and SHAP-based interpretability.

The proposed model achieves $R^2 = 0.936$, MAE=2.695 MPa, and RMSE=4.071 MPa on the UCI benchmark, outperforming both individual algorithms (including RF, ET, SVR, and MLP) and recent state-of-the-art stacking and boosting methods. The ablation study confirms stacking as the most impactful component; SHAP analysis reveals that cement content, curing age, and water-to-binder ratio are the dominant predictors, consistent with established concrete materials science.

Methodologically, the study demonstrates that leakage-free stacking with out-of-fold predictions and domain knowledge integration yields improved accuracy alongside interpretable, physically grounded predictions—supporting practical data-driven mixture design evaluation and reducing reliance on costly laboratory testing.

10 Reproducibility

All experiments used Python 3.10, scikit-learn 1.4.0, XGBoost 2.0.3, LightGBM 4.3.0, CatBoost 1.2.2, Optuna 3.5.0, SHAP 0.44.0, random seed 42.

Author contributions
Md. Siam Ansary is the sole author responsible for all aspects of this work.

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Data availability
The dataset analysed during the current study is publicly available in the UCI Machine Learning Repository (Concrete Compressive Strength dataset) at https://archive.ics.uci.edu/dataset/165/concrete+compressive+strength. All derived features used in this study were computed from the raw variables using the equations given in Sect. 3.3.

Declarations

Competing interests
The authors declare no competing interests.

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